

THE CONCEPT OF SPACE, TIME, AND CONSCIOUSNESS IN ANCIENT INDIA

ACADEMIC/SCIENTIFIC POSTER HINGUWA FEST-2024



Presented By: Nazong Nyu Lepcha & Bandana Chettri under the guidance of
Mr. Bivek Pradhan (Physics Department, Sikkim Govt Science College
Chakung)

Abstract

The concept of time, space & consciousness is not a new term it has been there since immemorial. Since ancient times the concept related to time and physics were known by the sages and ancient people and were mentioned in the manuscript like Puranas and Mahabharat. We quote mainly from Yoga-Vasistha (YV), which represents the thought of Indian in that period. This paper presents a speculation of recent physics also a brief account of the context in which ideas of YV developed.

KEY POINT-

- ❑ Ancient idea of Indian Physics, available to us through variety of sources, are generally not known in the physics world.
- ❑ There are two philosophical system at the basis of Indian Physics Samkhya and Vaishesika.
- ❑ Vedas are the text which represent the ancient knowledge tradition of India.
- ❑ Yoga-Vasistha (YV) describe the instruction given by Vasistha to RAMA.
- ❑ In ancient India time was cyclically rather than linearly, with concepts like kalpa. Space, also referred as akasha, which was seen as a fundamental elements from which all the matter and energy derived.

Methodology

In ancient Indian philosophy, the concepts of concepts of space, time and consciousness were explored in profound ways. Here's an overview

1) SPACE (Akasha):

- Vedic Perspective- In the Vedas, Akasha is considered the first element from which all the other elements arise.
- Philosophical Exploration: Later philosophical texts like the Upanishads delve deeper into the metaphysical aspects of akasha, portraying it as not just physical space but also as a symbol of the infinity.

2) TIME (Kala):

- Vedic Concept: Time in Vedic literature is multifaceted. It ranges from the cyclic time of the cosmos (yugas) to the ritualistic time (rita) and the divine (kala) controlled by gods.
 - Philosophical Interpretations: Philosophical schools such as Nyaya-Vaisheshika and Samkhya articulate time as a essential aspect of the cosmic order and individual experience.
- #### 2) CONSCIOUSNESS (Chitta):
- Vedic thought: It is a central in the Upanishads, where it is explored in terms of the individual self (Atman) and the universal consciousness (Brahman).

Vedic Cosmology: The methodology of vedic cosmology involves a blend of philosophical speculation, mythological narrative, observational astronomy. Here's a more detailed exploration:

- ❑ Philosophical Speculation: Vedic cosmology begins with fundamental questions about the nature of existence, the origin of the universe. It explores concepts such as creation (Srishti), sustenance (srithiti), and dissolution (laya) of the universe.
- ❑ Observational Astronomy: Early Vedic thinkers were keen observers of celestial phenomena. They studied the movements of the sun, moon, stars, and planets and moon incorporated their observations into cosmological models.

	VEDA	SCIENCE
Origin of universe	13.819 billion years ago	13.801 million years ago
Origin of sun	4.563 billion years ago	4.567 billion years ago

Puranic Cosmology: It follows a combine mythological Narrative with cosmological speculation.

- ❑ In puranas the universe is described as a radially different ways unlike the universe that is mentioned in today's world.
- ❑ In puranas there are several levels.
- ❑ At top we have Planetary system assembling a disk, at the middle we have earth and other planet and at lower we have asura lok.

The Yoga-Vasistha- It is structured as a dialogue between Sage Vasistha and Prince Rama, where Vasistha imparts spiritual wisdom to Rama. YV also describe samsara and reality as follows-
➤ Samsara is mundane existence with rebirth.
➤ The universe is full of samsara driven by Moha, bondage, malas.

Results

In ancient India, particularly within Hindu philosophical and spiritual tradition, there were profound explorations and insight.

1) Concept of Time:

- Cyclic Nature- Ancient Indian thought often viewed time as cyclical rather than linear. This concept is encapsulated in the idea of Yugas where the universe undergoes repeated cycles of creation, maintenance, dissolution, recreation.
 - Relativity of time- Time was understood to be relative, varying according to different planes of existence (lokas) and states of consciousness.
- #### 2) Concept of Space:
- Subtle essence- Akasha, often translated as space, is not merely physical but also represents the subtlest form of matter, pervading the entire cosmos. It is the medium through which everything manifests.
 - Multi-dimensional Cosmology – Ancient Indian texts describe a multi-layered cosmos with various realms existing within different levels of space. These realms are inhabited by celestial beings, gods, and other entities.

3) Concept of Consciousness:

- Unity and Diversity: Consciousness was seen as the fundamental substrate of existence, encompassing both the individual self (atman) and the universal consciousness (brahman). The unity underlies the diversity of manifest existence.
- Level of Consciousness: Ancient Indian philosophies classified consciousness into various states (waking, dreaming, deep sleep)
- And beyond ordinary awareness, highlighting the layers of awareness and perception.

THE YOGA-VASISTHA:

- The Yoga Vasistha is a profound philosophical text from ancient India attributed to the sages Valmiki. It explores the themes of consciousness, existence, and the nature of reality through a series of dialogues between Prince RAMA and his Guru Vasistha.
- The text is notable for its detailed explanation of Advaita Vedanta, which teaches that the ultimate reality is non-dual consciousness
- In ancient India, the Yoga Vasistha had a significant influence on Hindu philosophy and spiritual thought. It provided deep insights into the nature existence and offered guidance on how to attain liberation through self realization.

Results of Vedic and Puranic Cosmology:

Vedic Cosmology-

- The Rigveda, one of the oldest vedic text presents a cosmology where the universe is conceptualized in layers. It describes the earth as a flat disc surrounded by the sky.
- Vedic hymns often depict celestial bodies and natural phenomena as divine forces. Rituals were performed to maintain cosmic order and harmony.
- Vedic cosmology emphasizes the principle of ritam, representing the natural order and cosmic laws governing the universe.

Puranic Cosmology-

- Puranas such as the Vishnu purana and Bhagavada purana elaborate on creation myths where the universe emerges cyclically through the actions of divine beings like Bramha, Vishnu, Shiva.
- Puranic text describe the universe as a multilayered cosmos with realms including higher realms like Swarga, and lower realms like Patala, along with the earthly realm.

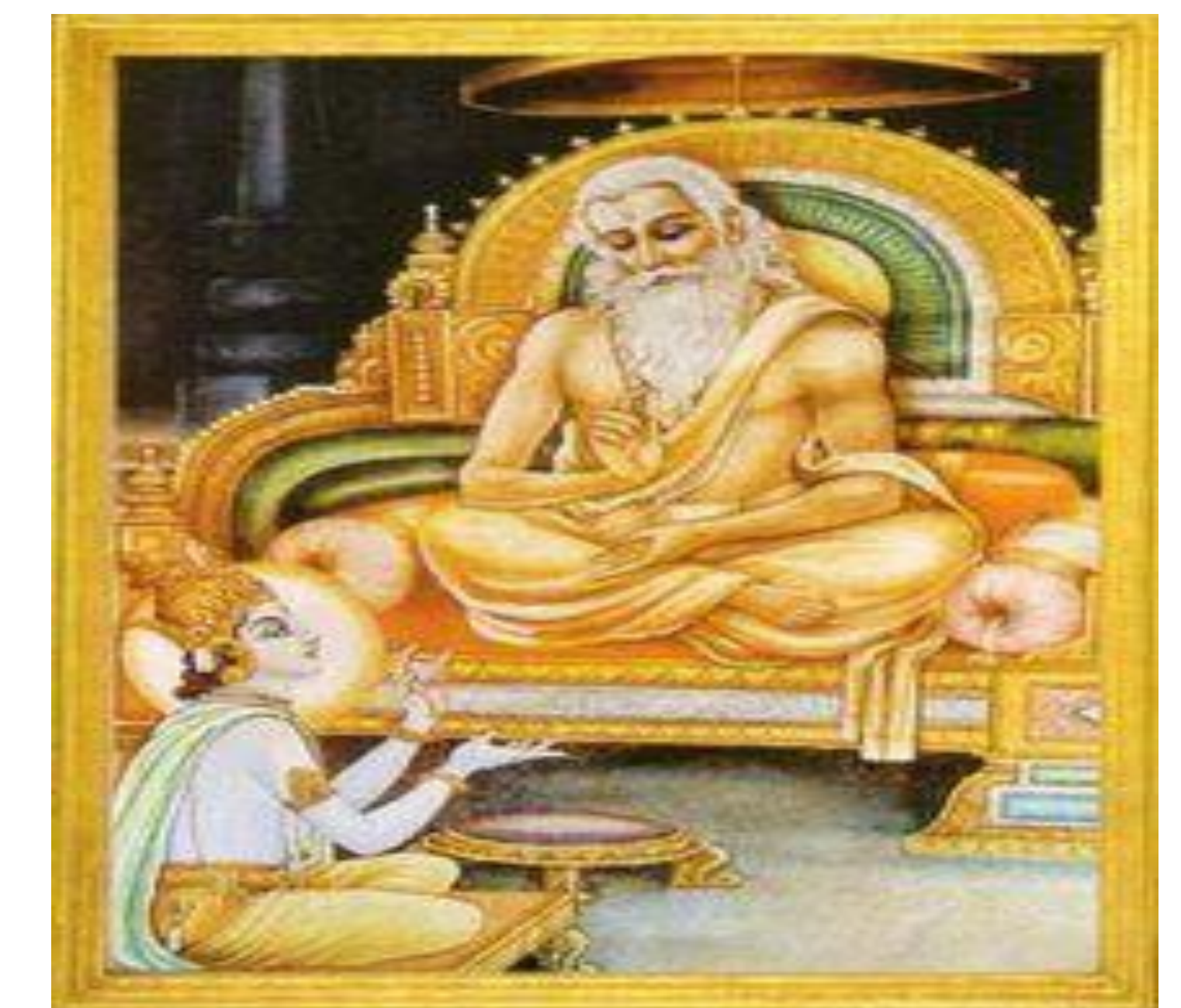
Conclusion

It appears that Indian understanding of physics was informed not only by astronomy and terrestrial experiments but also by speculative thought and by meditations on the nature of consciousness. Unfettered by either geocentric or anthropocentric views, this understanding unified the physics of the small with that of the large within a framework that included metaphysics. This was a framework consisting of innumerable worlds (solar systems), where time and space were continuous, matter was atomic, and consciousness was atomic, yet derived from an all-pervasive unity. The material atoms were defined first by their subtle form, called tanmatra, which was visualized as a potential, from which emerged the gross atoms. A central notion in this system was that all descriptions of reality are circumscribed by paradox (Kak,1986).The universe was seen as dynamic, going through ceaseless change.

YOGA-VASISTHA:

- The Yoga Vasistha continues to be studied and reversed by seekers of spiritual wisdom and scholars interested in Indian philosophy. Its impact can be seen in various spiritual traditions and philosophical discourses throughout history.

Amongst popular books, Paul Halpern's The Cyclical Serpent (1995) is unusual in that it places modern speculations regarding an oscillating universe within the context of the cyclic cosmology of the Puranas, but even this book doesn't define a context for the Indian ideas. In this paper we present, in a capsule form, the basic Indian ideas on space, time, and observation from the age of the epics and the early puranas.



References

- P. Halpern, 1995. The Cyclical Serpent: Prospects for an Ever-Repeating Universe. Plenum Press, New York.
- S. Kak, 1986. The Nature of Physical Reality. Peter Lang, New York.
- S. Kak, 1990. Kalidasa and the Agnimitra problem. Journal of the Oriental Institute 40: 51-54.
- S. Kak, 1994. The Astronomical Code of the R. gveda. Aditya, New Delhi.
- S. Kak, 1995a. From Vedic science to Vedānta. Brahmagyāna: The Adyar Library Bulletin, 59: 1-36.
- S. Kak, 1995b. The astronomy of the age of geometric altars. Quarterly Journal of the Royal Astronomical Society 36: 385-396.
- S. Kak, 1996. Knowledge of planets in the third millennium BC. Quarterly Journal of the Royal Astronomical

Introduction

- In ancient India, the exploration of space, time, and consciousness was intricately woven into philosophical and spiritual discourse. These concepts were not merely intellectual pursuits but formed the bedrock of profound insight into the nature of existence and the human experience.
- The information in the scriptures shares a similarity with today's physics.
- The most fascinating numerical claims from the ancient Indians are-
 - 1) A cyclic system of creation of the universe with the period of 8.64 billion years, there might exist longer cycles as well.
 - 2) Speed of light to be 4,404 yojanas per nimesa, which is almost exactly 186,000 miles per second.
- Ancient India has two philosophical systems on the basis of Physics & Metaphysics: Samkhya & Vaishesika

1) Samkhya- It has 25 basic categories & 3 constituent qualities which evolve in different ways to create universe at microscopic & macroscopic levels.

- Samkhya makes a distinction

Between two "irreducible, innate and independent realities", Purusha, the witness Consciousness, and Prakriti, "matter, the activities of mind and Perception.

2) Vaishesika- It is a system of atomic theory with non-atomic ground. It considers mind to be atomic.

